28.05.2025 (E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: January 2025 –April 2025		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01 Programme: B. Tech. COMP	Class: TY	Semester: V(SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMP/~~ETRX/EXTC/IT/MECH/RAI/AIDS/CCE/EXCP~~	
Course Code: 116U01C503	Name of the Course: Operating System	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **FOUR**	20
i)	What is operating system? Explain multiprogramming and time-sharing systems.	5
ii)	Write a shell program to reverse a given number, and check whether it is palindrome or not.	5
iii)	Describe the purpose and components of a resource allocation graph (RAG) and explain how it helps in detecting deadlocks within a system.	5
iv)	List the various file access methods. Explain any one in detail with advantage.	5
v)	Explain the concepts of dynamic linking and dynamic loading in an operating system. How do these mechanisms improve memory efficiency and program execution?	5
vi)	Distinguish between the short-term scheduler and the medium-term scheduler	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Explain Linux's Completely Fair Scheduler (CFS).	5
ii)	Describe the key components of a Process Control Block (PCB) and analyse its role in the efficient management of processes within an operating system.	5
	OR	
Q2 A	For the given set of processes, apply the **Shortest Remaining Time First (SRTF) and Pre-emptive Priority** Scheduling algorithms. Note: Lower number indicates higher priority.	10

Process	Arrival Time (ms)	Burst Time (ms)	Priority
P1	0	5	1
P2	1	3	2
P3	2	8	1
P4	3	6	3

a) Construct the Gantt chart for each algorithm.
b) Compute the Average Waiting Time and Turnaround Time for the processes under each scheduling method.

Q 2 B	Solve any **ONE**	10
i)	Provide a semaphore-based solution for the Reader-Writer Problem by: a) Writing the pseudo-code for the solution. b) Demonstrating how semaphores prevent deadlock and starvation.	10
ii)	Provide a monitor-based solution for the Dining Philosophers Problem by: a) Writing the pseudo-code for the solution. b) Demonstrating how monitor prevent- deadlock and starvation.	10

Que. No.	Question	Max. Marks
Q3	Solve any **TWO**	**20**
i)	For the reference string {7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1} and frame size of 3, demonstrate FIFO & LRU Page Replacement with steps.	10
ii)	Explain the concepts of hierarchical page table and inverted page table in detail, highlighting their structure, working mechanism, and differences.	10
iii)	What is demand paging? Explain it with address translation mechanism used. What are its specific advantages? How a page table is implemented?	10

Que. No.	Question	Max. Marks
Q4	Solve any **TWO**	**20**
i)	Brief explanation of hash and clustered file organization, along with their advantages and disadvantages. Illustrate with suitable diagram.	10
ii)	Consider, a disk queue with requests for I/O to blocks on cylinders in the following order: 98,183,37,122,14,124,65,67 The disk head pointer is initially at cylinder 53. Outline Shortest Seek Time First (SSTF) and SCAN Disk scheduling algorithm. a) Sketch the diagram for both algorithms. b) Calculate the seek time for both algorithms.	10
iii)	Consider the following snapshot of a system at t:	10

Process	Allocation				Max			
	A	B	C	D	A	B	C	D
T0	0	0	1	2	0	0	1	2
T1	1	0	0	0	1	7	5	0
T2	1	3	5	4	2	3	5	6
T3	0	6	3	2	0	6	5	2
T4	0	0	1	4	0	6	5	6

Available			
A	B	C	D
1	5	2	0

Using the banker's algorithm:

a) What is the content of the matrix need? (02M)
b) Is the system in a safe state? (05M)
c) If a request from thread T1 arrives for (0,4,2,0) can the request be granted immediately. (03M)

Que. No.	Question	Max. Marks
Q5	Solve any **FOUR**	**20**
i)	Write short note on queuing diagram. w.r.to 5-state Transition diagram	5
ii)	Write Short note on Linux Virtual File System.	5
iii)	Explain safe, unsafe and deadlock state process with the help of example.	5
iv)	What is virtual memory? Discuss the benefits of virtual memory techniques.	5
v)	Define a compiler and discuss its significance in a computing system, highlighting its key functions and impact on program execution.	5
vi)	Discuss the direct and indirect communication strategies in message-passing systems, with help of example.	5